

Dante Buhl

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EDUCATION

University of California, Santa Cruz

- **Ph.D., Applied Mathematics** — Graduate GPA 4.0/4.0 · 2024–Present (expected 2028) · Advisor: Prof. Pascale Garaud
- **M.S., Scientific Computing & Applied Mathematics** — GPA 4.0/4.0 · 2024
- **B.A., Mathematics, Highest Honors (Computational Mathematics)** — GPA 3.74/4.0 · 2023

RELEVANT COURSEWORK

Stochastic Differential Equations (A) · Probability Theory (A–) · Numerical Linear Algebra (A+) · Numerical Methods for Differential Equations (A+) · Applied & Advanced PDEs (A+) · High Performance Computing (A+) · Applied Dynamical Systems (A) · Scientific Machine Learning (A) · Computational Fluid Dynamics (A) · Advanced Linear Algebra (A) · Complex Analysis (B+)

TECHNICAL SKILLS

Programming: Python, Fortran, MATLAB, R, Java, JavaScript

Quantitative: Stochastic differential equations & methods, probability, numerical linear algebra, PDEs, time-series & dynamical systems, Monte Carlo & numerical simulation, statistical / scientific machine learning

Computing: MPI parallel programming, high-performance & cluster computing, $\text{\LaTeX}$

RESEARCH EXPERIENCE

Graduate Research — UC Santa Cruz, Applied Mathematics

Sept 2023–Present

- Build and run large-scale MPI-parallel simulations (Fortran) of rotating stratified turbulence on HPC clusters, modeling high-dimensional, nonlinear stochastic dynamics.
- Extract statistical scaling laws and transport coefficients from large simulation datasets, quantifying how rotation governs turbulent mixing.
- Apply SDEs, spectral / finite-difference solvers, and numerical linear algebra to multiscale PDE systems; presented results at two international conferences (APS DFD 2024, EGU 2025).

Undergraduate Research — UC Santa Cruz, Mathematics

2023

- “Boxes and Balls: Quantifying Chaos”: developed numerical methods to compute the fractal dimension of chaotic attractors.

Research Assistant — Towson University Mathematics REU

Summer 2023

- Developed machine-learning-based numerical solvers (Python) for the Navier–Stokes equations in low-Reynolds (Stokes) flow.

PROJECTS

Statistical Arbitrage via Ornstein–Uhlenbeck Mean Reversion

Fall 2026

- Modeling cointegrated equity-pair spreads as Ornstein–Uhlenbeck processes; deriving the exact discretization and estimating mean-reversion speed, half-life, and volatility by maximum likelihood (Python).
- Selecting pairs with Engle–Granger cointegration tests and multiple-testing controls; tracking time-varying hedge ratios with a Kalman filter.
- Backtesting z-score entry/exit rules strictly out of sample with transaction costs; reporting Sharpe ratio, drawdown, and turnover, and diagnosing periods where pair relationships break down.

TEACHING & LEADERSHIP

- **Teaching Assistant, UC Santa Cruz (2023–Present):** Dynamical Systems (AM 114), Computational Methods (AM 147), PDEs (AM 112), Mathematical Methods for Economists (AM 11B), Calculus I–II (Math 19A/B).
- **Secretary, Slugs United by Mathematics** (2022–2024).

AWARDS

Next Generation Scholars in Applied Mathematics (2023) · T&LC Graduate Pedagogy Fellowship · B.A. with Highest Honors